VGA maze display on the PYNQ-Z2

Digital Design Concepts — Faculty of Engineering Technology, KU Leuven

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1 PROJECT OVERVIEW

We built an interactive VGA display on a PYNQ-Z2 board using Verilog and Xilinx Vivado. Four push buttons move a red square horizontally, vertically or diagonally, while position limits keep it within the visible screen. We also added a blue maze as a graphical setting. **Figure 1** shows the display and board during the project.

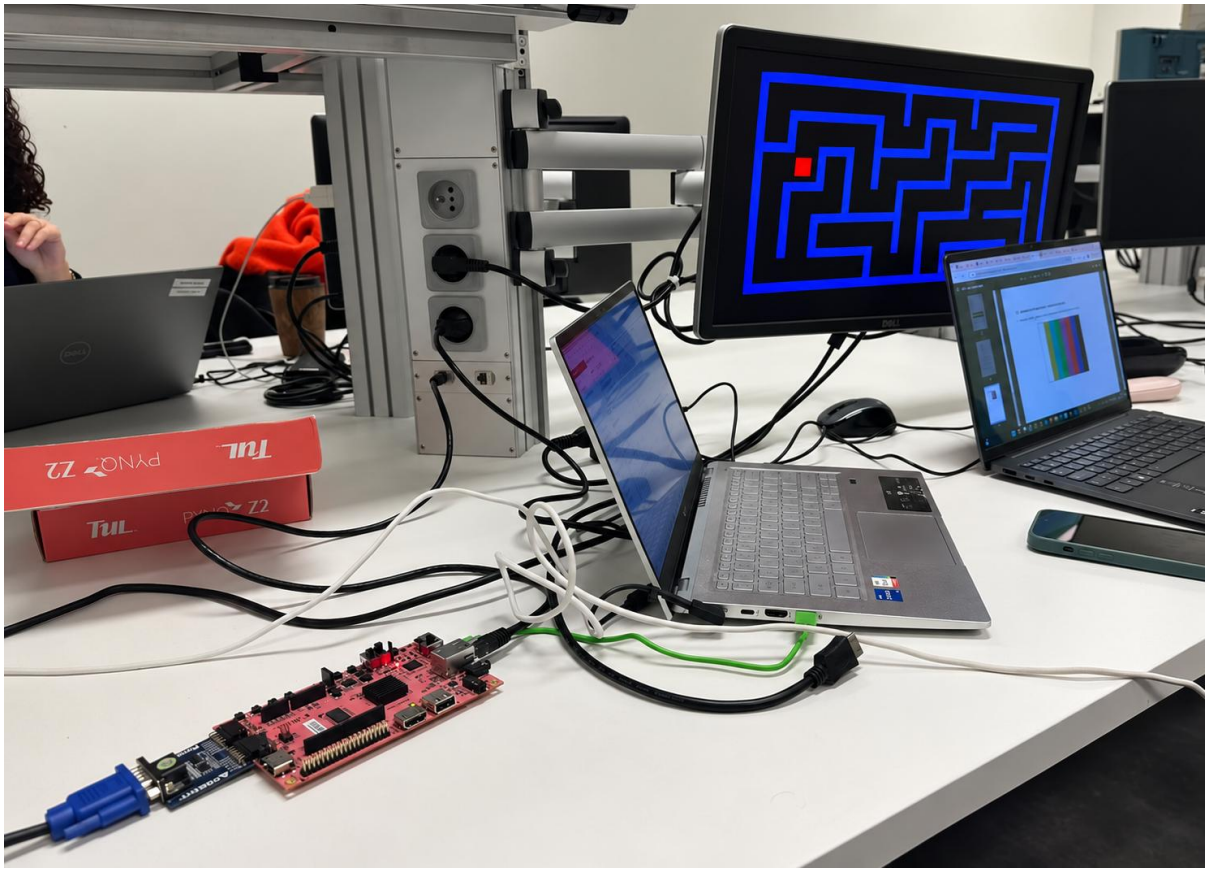


Figure 1: The project running on the PYNQ-Z2: the monitor displays the blue maze and red square, with the board and VGA connection visible in the foreground.

2 DESIGN AND IMPLEMENTATION

The design separates button handling, position updates and video generation. Debounce modules condition the inputs before the movement logic uses them. This logic stores the square's coordinates and size. VGA timing and pattern modules generate the synchronisation signals and pixel colours, with a clock-generation block supplying the video logic.

The finite-state machine has three states: sinit, sidle and sMove. After reset it enters sinit and then advances automatically to sidle. A directional input selects sMove. The machine stays in sMove while an input remains active and returns to sidle once all directional inputs are released.

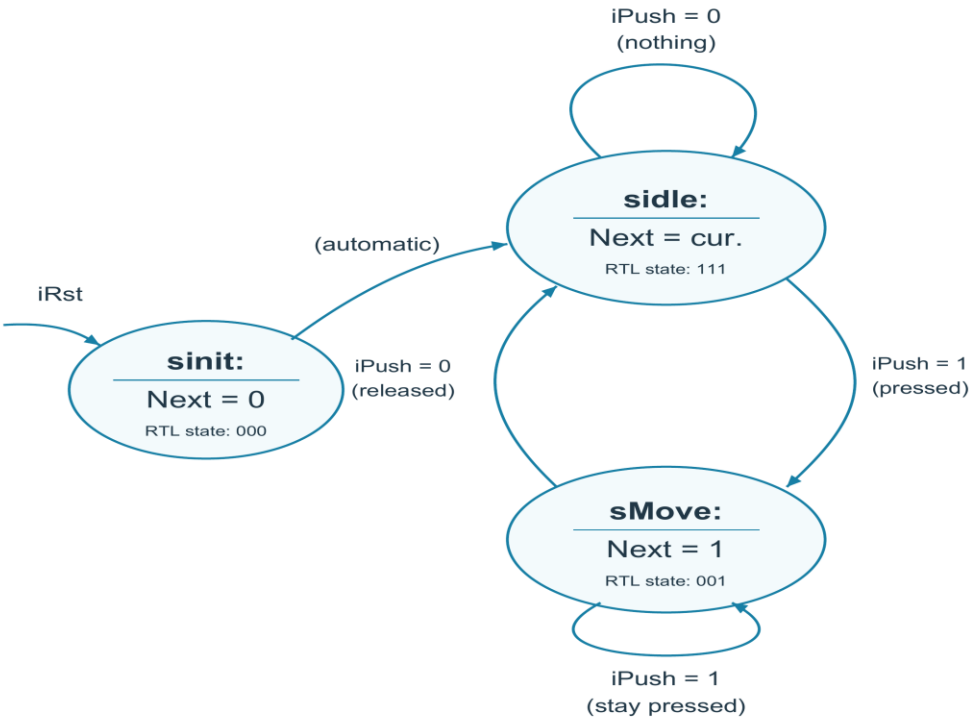


Figure 2: Movement control FSM. The Next labels follow the project sketch; the RTL state encodings are shown below them. iPush represents a directional input. Reset returns the machine to sinit.

3 USER INTERACTION AND VISUAL OUTPUT

Horizontal and vertical inputs are evaluated independently. Pressing one vertical and one horizontal button therefore produces diagonal movement. Holding a button continues the movement; releasing the buttons keeps the square at its current position.

Table 1: Mapping of each button press / input into specific monitor actions

Control	Board input	Action
Up / Down	BTN0 / BTN1	Move vertically
Left / Right	BTN2 / BTN3	Move horizontally
Diagonal	One vertical and one horizontal button	Update both coordinates
Reset	SW0	Restore the initial position and state

The square is 20 × 20 pixels and starts at X = 290, Y = 210, near the middle of the 640 × 480 image. The position limits account for the square's size so that it stays on screen. The graphics module draws the square in red, the maze walls in blue and the remaining background in black. The wall layout is defined by coordinate ranges.

The board LEDs provide a small visual indication of button activity. A separate LED-feedback video accompanies the project materials.

4 IMPLEMENTATION RESULTS

The project targets the xc7z020clg400-1 device and uses the PYNQ-Z2 pin constraints. The Vivado 2020.1 implementation run completed bitstream generation on 18 December 2024, producing design_1_wrapper.bit. The routed timing report confirms that all user-specified timing constraints were met.

5 CONCLUSION

The project combines push-button control, a movement FSM and VGA graphics on one FPGA. It provides eight-direction movement, screen-boundary control and LED feedback, with the maze forming the visual setting. The photograph records the display output, while the separate video shows the LED feedback.